Brazilian E-Commerce

Business Analysis Report

Olist Platform • SQL Analytics • 2016–2018

Prepared by: Anupam Kumar | Tools: MySQL Workbench, Advanced SQL

1. Executive Summary

This report presents a structured analysis of Olist, Brazil's leading e-commerce marketplace, covering 96,461 orders placed by 99,441 customers between September 2016 and September 2018. Using Advanced SQL techniques - including window functions, CTEs, and composite ranking - twelve business questions were answered across revenue trends, customer behaviour, operational performance, and seller quality.

Three findings stand out above all others:

- Revenue is growing but seasonal - Q4 spikes are predictable and should be resourced in advance, while mid-year dips represent an opportunity for targeted promotional campaigns.
- The repeat buyer is rare but disproportionately valuable - only 2.9% of customers return for a second purchase, yet these customers generate a majority of High-Value segment revenue. Retention investment here yields the highest ROI.
- Delivery SLA breaches affect approximately 8–10% of orders - concentrated among specific sellers and product categories - directly damaging customer satisfaction and review scores.

These insights equip Marketing, Operations, Finance, and Product teams with data-driven foundations for 2024 planning decisions.

2. Dataset Overview

2.1 Tables & Row Counts

Table	Rows	Description
olist_orders	96,461	Order lifecycle: status, timestamps, delivery dates
olist_customers	99,441	Customer IDs, zip codes, state
olist_order_items	112,650	Line items: product, seller, price, freight

olist_order_payments	103,886	Payment type, instalments, value per order
olist_order_reviews	104,719	Star ratings and review text per order
olist_products	32,951	Product dimensions, weight, category name (PT)
product_category_names	71	Portuguese-to-English category translations

2.2 Key Data Discovery: customer_id vs customer_unique_id

A critical discovery early in the analysis: Olist's schema assigns a new customer_id for every order, meaning that a single real-world customer who places three orders will appear as three different records in the customers table. The customer_unique_id field is the true persistent identity key. Without this distinction, repeat purchase rates, customer lifespan, and inter-purchase gap analyses would be completely invalid.

- customer_id: order-scoped - resets with each new order (cannot be used for cohort analysis)
- customer_unique_id: customer-scoped - stable across all orders by the same person
- Repeat buyers identified: 2,800 of 99,441 unique customers (~2.9%)

2.3 Scope: Delivered Orders Only

All revenue and delivery analyses are restricted to orders with order_status = 'delivered'. Cancelled, invoiced, and processing orders were excluded to ensure revenue figures reflect actual completed transactions and to make delivery SLA comparisons valid. This constraint reduces the effective order set slightly but significantly improves analytical accuracy.

3. Analysis & Insights

Q1 - Monthly Revenue Trend

Business Question	How has revenue trended month over month?
SQL Logic	Uses a CTE to aggregate monthly revenue, then applies the LAG() window function to compare each month's revenue against the previous month and calculate a percentage change.
Key Insight	Revenue shows strong seasonal spikes in late Q4 (November–December), consistent with Brazilian holiday shopping. Mid-year months (June–July) show recurring dips, suggesting predictable low seasons.
Business Decision	Enables Marketing to plan promotional budgets in advance of peak months and Finance to set realistic monthly targets against historical baseline.
Team Benefit	Marketing · Finance

Q2 - Cumulative Revenue Growth

Business Question	What is cumulative revenue growth over time?
SQL Logic	Applies SUM() as a running total window function (ORDER BY month, ROWS UNBOUNDED PRECEDING) to accumulate revenue from the first data point forward.
Key Insight	Cumulative revenue grew steadily from ~2016 through a steep acceleration in 2017–2018, confirming strong platform-level growth momentum rather than one-time spikes.
Business Decision	Gives the Finance team a compounding revenue baseline for investor reporting and helps set credible multi-year growth targets.
Team Benefit	Finance · Executive

Q3 - Top Product Categories by Revenue

Business Question	Which categories drive the most revenue?
SQL Logic	Joins order items with translated product categories, aggregates revenue per category, and uses RANK() with a CROSS JOIN to calculate each category's percentage share of total revenue.
Key Insight	Health & Beauty, Watches & Gifts, and Bed/Bath/Table consistently rank in the top three, together accounting for a significant share of gross merchandise value.
Business Decision	Product and Marketing teams can concentrate promotional spend and inventory investment on proven revenue-generating categories, and can identify underperforming categories for review.
Team Benefit	Product · Marketing

Q4 - Customer Value Segmentation

Business Question	How do we classify customers into High / Mid / Low value tiers?
SQL Logic	Uses NTILE(4) to divide customers into quartiles by lifetime spend, then CASE WHEN to label them High, Mid-High, Mid-Low, and Low value segments.
Key Insight	The top quartile (High-Value) accounts for a disproportionate share of total revenue despite representing only 25% of the customer base. Critically, nearly all High-Value buyers have made more than one purchase.
Business Decision	CRM can build targeted loyalty and retention programmes exclusively for the High-Value tier, where retention ROI is highest.
Team Benefit	Marketing · CRM

Q5 - Day-of-Week Order Patterns

Business Question	Which days see highest order volume for campaign scheduling?
SQL Logic	Extracts the day of week from the order timestamp using DAYOFWEEK(), groups orders by day, and ranks days by total volume.
Key Insight	Monday and Tuesday consistently rank as the highest-volume order days. Weekends see notable drops, suggesting customers browse during the week rather than on days off.
Business Decision	Marketing should schedule major email campaigns and push notifications for Sunday evening dispatch to land on Monday morning, aligning with natural buying behaviour.
Team Benefit	Marketing

Q6 - Order Volume Forecast

Business Question	What does next month's volume look like vs current?
SQL Logic	Uses both LAG() and LEAD() window functions to place the previous and next month's order counts side-by-side in a single row, enabling direct comparison without self-joins.
Key Insight	Volume spikes are largely predictable 4–6 weeks in advance based on prior-year patterns and the rolling trend, giving Operations a reliable short-term planning signal.
Business Decision	Warehouse and logistics teams can pre-position stock and scale staffing approximately 6 weeks before forecast volume peaks.
Team Benefit	Operations · Supply Chain

Q7 - Customer Lifespan Analysis

Business Question	First vs latest purchase - how long do customers stay active?
SQL Logic	Uses ROW_NUMBER() to identify each customer's first and last order, then DATEDIFF() to measure the span between them. NULLIF() prevents divide-by-zero on single-purchase customers.
Key Insight	The majority of customers have an active window of fewer than 45 days, reflecting the low 2.9% repeat purchase rate. Genuine long-lifespan customers are a small but extremely valuable cohort.
Business Decision	The 45-day mark is a natural trigger point for re-engagement automations - customers who have not reordered by Day 30 should receive targeted win-back messaging.
Team Benefit	CRM · Marketing

Q8 - Rolling 3-Month Revenue Average

Business Question	Is revenue growth consistent or volatile?
SQL Logic	Applies a sliding window average using SUM() OVER (ORDER BY month ROWS BETWEEN 2 PRECEDING AND CURRENT ROW) to smooth monthly revenue variance across a 3-month window.
Key Insight	Month-to-month revenue is volatile (±20–30% swings), but the smoothed 3-month average shows a steady upward trend, indicating that individual spikes and dips are noise rather than signal.
Business Decision	Finance should adopt the rolling 3-month average as the primary KPI for revenue health reporting, rather than relying on single-month comparisons that can mislead stakeholders.
Team Benefit	Finance · Executive

Q9 - Payment Method by Customer Segment

Business Question	Do high-value customers pay differently than low-value ones?
SQL Logic	Combines NTILE() segmentation from Q4 with PARTITION BY payment_type to show the distribution of payment methods within each customer value tier.
Key Insight	High-value customers show a stronger preference for credit card instalments (parcelas), while low-value buyers skew toward single-instalment or boleto payments - reflecting differences in purchase size and financial access.
Business Decision	Product can prioritise instalment credit options in the checkout UX for returning or identified high-value customers, reducing friction that might suppress larger basket sizes.
Team Benefit	Product · Finance

Q10 - Delivery SLA Breach Analysis

Business Question	Which orders were late and what % breach the SLA?
SQL Logic	Uses DATEDIFF() to compare actual delivery date against estimated delivery date, then CASE WHEN to flag breaches. Results are aggregated by seller and product category to identify systemic issues.
Key Insight	Approximately 8–10% of delivered orders arrive after the estimated date. Breaches cluster around specific sellers and product categories with longer fulfilment times (e.g. furniture, large appliances).
Business Decision	Operations can implement a seller SLA scorecard; sellers with breach rates above a threshold should receive performance improvement notices or revised estimated delivery windows.
Team Benefit	Operations · Logistics

Q11 - Inter-Purchase Gap Analysis

Business Question	How many days do repeat buyers wait between purchases?
SQL Logic	Filters for customers with more than one order, uses LAG() to align consecutive purchases in the same row, then DATEDIFF() to calculate the gap between each pair.
Key Insight	Among repeat buyers, the median inter-purchase gap is approximately 60–90 days, with a significant cluster around 30 days. This indicates a natural replenishment or consideration cycle.
Business Decision	CRM teams can configure automated re-engagement sequences to fire at Day 30 and Day 60 post-purchase, timed to align with the observed buying rhythm.
Team Benefit	CRM · Marketing

Q12 - Seller Composite Performance Ranking

Business Question	Who are the top sellers across revenue, volume, and review score?
SQL Logic	Independently ranks sellers on three dimensions (total revenue, order volume, average review score), then constructs a weighted composite score and applies a final RANK() to produce a blended leaderboard.
Key Insight	The top 10% of sellers by composite score contribute disproportionately to both revenue and customer satisfaction. A subset of high-revenue sellers carry below-average review scores, flagging a quality/scale tension.
Business Decision	Seller Success teams can use the composite ranking to identify and reward true top performers, and to proactively support high-volume sellers whose review scores are dragging down customer experience.
Team Benefit	Seller Success · Operations

4. Cross-Cutting Themes

Theme 1: The Repeat Buyer is Rare but Disproportionately Valuable

Across four separate analyses - customer value segmentation (Q4), customer lifespan (Q7), inter-purchase gap (Q11), and payment behaviour (Q9) - the same small cohort of ~2,800 repeat buyers surfaces repeatedly as the engine of sustainable revenue. These customers have longer active windows, higher basket values, and a preference for credit instalments that enables larger single-transaction spend. The business implication is stark: the cost of acquiring a new customer far exceeds the cost of retaining an existing one, yet the platform's current structure is overwhelmingly transactional and single-purchase. Every team - from Marketing to Product to CRM - has both the data and the incentive to shift strategy toward retention.

Theme 2: Predictability is an Underused Asset

The monthly revenue trend (Q1), rolling average (Q8), and volume forecast (Q6) analyses collectively reveal that Olist's growth pattern is more predictable than day-to-day volatility suggests. Q4 holiday spikes, mid-year dips, and week-day buying patterns all follow consistent historical rhythms. This predictability is currently being wasted: promotional budgets are not clearly aligned to peak periods, warehouse staffing appears reactive rather than proactive, and campaign scheduling does not reflect the Monday-heavy order pattern. Turning this latent predictability into operational planning inputs is one of the highest near-term value opportunities.

Theme 3: Quality and Scale Create Opposing Pressures on Sellers

The seller composite ranking (Q12) and delivery SLA analysis (Q10) together reveal a structural tension in Olist's seller ecosystem. High-volume sellers frequently carry below-average review scores and elevated delivery breach rates, while smaller sellers who manage quality carefully cannot scale without operational support. This tension, if left unaddressed, will erode platform reputation as GMV grows. A two-track seller support model - one focused on scaling quality sellers and one focused on repairing the quality of high-volume sellers - is indicated by the data.

Theme 4: Payment Behaviour as a Proxy for Customer Intent

The payment method analysis (Q9) illuminates a dimension of customer intent that is rarely surfaced in e-commerce reporting. High-value customers do not simply spend more - they structure their spending differently, using instalment credit to manage larger purchases. This has direct product implications: checkout UX, instalment offer visibility, and credit partner integrations are not just financial decisions but customer experience decisions that affect whether the platform's best customers return. Combined with the inter-purchase gap analysis (Q11), this suggests that reducing friction at checkout for identified repeat buyers could meaningfully compress the 60–90 day return cycle.

5. Recommendations Summary

Analysis Area	Key Finding	Recommended Action	Team Owner
Monthly Revenue Trend	Q4 peaks; mid-year dips are consistent	Plan inventory & ad spend 6–8 weeks before peak	Marketing / Finance
Cumulative Revenue	Steady compounding growth across 2 yrs	Set annual revenue targets anchored to trend line	Finance
Top Categories by Revenue	Health & Beauty, Watches lead revenue	Prioritise these categories for promotions & shelf space	Product / Marketing
Customer Value Segmentation	Only ~3% are repeat, High-Value buyers	Build loyalty programme targeting High-Value tier	Marketing / CRM

Day-of-Week Patterns	Mon–Tue carry highest order volume	Schedule email campaigns for Sunday night / Monday AM	Marketing
Order Volume Forecast	Volume spikes are predictable 4–6 wks out	Pre-position warehouse stock before forecast peaks	Operations
Customer Lifespan	Avg active window < 45 days for most buyers	Trigger re-engagement emails at Day 30 post-purchase	CRM / Marketing
Rolling 3-Month Avg	Revenue is volatile; smoothed trend is upward	Use 3-month avg as KPI baseline, not single-month	Finance
Payment Method by Segment	High-value buyers prefer instalment credit	Offer instalment options prominently at checkout	Product / Finance
Delivery SLA Breach	~8–10% of orders breach estimated delivery	Audit top-breach seller & carrier combos immediately	Operations / Logistics
Inter-Purchase Gap	Repeat buyers return every 60–90 days avg	Set win-back automation at Day 60 after last order	CRM / Marketing
Seller Performance Ranking	Top 10% sellers drive disproportionate revenue	Offer elite sellers priority support & fulfilment SLA	Ops / Seller Success

6. Limitations & Assumptions

6.1 Data Period Constraint

The dataset covers September 2016 to September 2018 - a two-year window that includes Olist's early growth phase. Trends identified here may not reflect current platform behaviour, particularly given Brazil's subsequent economic cycles and the growth of domestic e-commerce competition post-2018.

6.2 Same-Day Purchase Filtering

For inter-purchase gap analysis (Q11), orders placed on the same calendar day by the same customer were treated as a single purchase event to avoid artificially inflating the frequency of repeat buyers. This conservative assumption may slightly undercount true repeat instances.

6.3 Currency & Regional Context

All monetary values are in Brazilian Reais (BRL). No currency conversion was applied. Revenue figures should not be directly compared to international benchmarks without adjustment. Additionally, Olist serves a geographically diverse market - state-level variation in delivery times, payment preferences, and product demand is present in the data but not fully disaggregated in this analysis.

6.4 What Additional Data Would Improve This Analysis

- Customer acquisition channel (organic, paid, referral) - essential for full CAC/LTV modelling

- Product return and refund records - needed to calculate net revenue rather than gross
- Marketing spend by campaign - would enable ROI attribution by channel and period
- Seller operational data (warehouse location, fulfilment partner) - would strengthen SLA breach root-cause analysis
- Session and clickstream data - would allow funnel analysis and cart abandonment measurement

7. SQL Skills & Tools Demonstrated

SQL Concept	Used In	Why It Was Chosen
Window Functions (LAG, LEAD)	Q1, Q6, Q11	Avoids expensive self-joins; enables row-to-row comparison within a single pass over sorted data
Running Total (SUM OVER)	Q2, Q8	Accumulates values without collapsing the result set, preserving row-level granularity alongside aggregates
NTILE()	Q4, Q9	Creates equal-sized customer tiers without requiring manual threshold setting, making segmentation reproducible
RANK() / DENSE_RANK()	Q3, Q12	Assigns competitive rankings with tie-handling, critical for leaderboard and category-share analysis
CTEs (WITH clause)	Q1, Q7, Q12	Improves readability and logical modularity; allows intermediate result sets to be referenced by name
DATEDIFF + CASE WHEN	Q10, Q11	Converts date arithmetic into categorical labels (e.g. On Time vs Breached), enabling aggregation of time-based outcomes
CROSS JOIN	Q3	Pairs every category row with a single-row total to calculate percentage share without a subquery in SELECT
ROWS BETWEEN sliding window	Q8	Implements a precise 3-month rolling average that moves with each row, more accurate than calendar-based grouping
ROW_NUMBER() + NULLIF()	Q7	Identifies first/last events per partition and safely handles division where a customer has only one order
Weighted Composite Score	Q12	Enables multi-dimensional ranking by normalising disparate metrics (revenue, volume, score) into a single comparable index

Addendum: Professional Assets

A. Resume Project Description

> *Conducted end-to-end SQL analysis of Olist's Brazilian e-commerce dataset (99K+ customers, 96K orders, 2016–2018) using MySQL Workbench. Applied advanced techniques - window functions, CTEs, NTILE segmentation, rolling averages, and composite ranking - across 12 business questions spanning revenue trends, customer value, delivery SLA, and seller performance.*

B. LinkedIn Post

> I just finished a deep-dive SQL analysis of a real e-commerce dataset - and the data told me something that changed how I think about customer strategy.

Out of 99,441 customers on Brazil's Olist platform, only 2,800 - less than 3% - ever came back for a second purchase. But those 2,800 people? They drive an outsized share of total revenue, have longer basket values, and use credit instalments to make bigger purchases. The data is clear: the real growth lever is not acquisition. It is retention. Using MySQL Workbench, I answered 12 business questions using advanced SQL: window functions (LAG, LEAD, NTILE, RANK), CTEs, rolling averages, delivery SLA breach detection, and a weighted composite seller scorecard. The kind of analysis that turns raw transaction logs into decisions a CMO or COO can act on tomorrow. If you are hiring analysts who can do this - or if you want to swap notes on window functions - let us connect. #SQLAnalytics #BusinessIntelligence #DataAnalysis #EcommercePlatform #PortfolioProject

C. 30-Second Interview Pitch

> "I recently completed a full SQL analytics project on Olist, a Brazilian e-commerce platform, with nearly 100,000 customers and two years of transaction data. I answered 12 business questions - everything from monthly revenue trends and customer value segmentation to delivery SLA breach detection and seller performance ranking. The most interesting finding was that less than 3% of customers ever make a second purchase, but that small group drives a disproportionate share of revenue - which has direct implications for where a business should invest in retention vs acquisition. I used advanced SQL throughout: window functions like LAG, LEAD, NTILE and RANK, CTEs for modular logic, rolling averages for trend smoothing, and a weighted composite score for multi-dimensional seller ranking. The whole project was about turning a raw dataset into decisions a marketing director, ops lead, or CFO could act on - which is exactly what I aim to do in this role."